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On the Class of Risk Neutral Densities under Heston's Stochastic Volatility Model for Option Valuation

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Abstract: The celebrated Heston's stochastic volatility (SV) model for the valuation of European options provides closed form solutions that are given in terms of characteristic functions. However, the numerical calibration of this five-parameter model, which is based on market option data, often remains a daunting task. In this paper, we provide a theoretical solution to the long-standing 'open problem' of characterizing the class of risk neutral distributions (RNDs), if any, that satisfy Heston's SV for option valuation. We prove that the class of scale parameter distributions with mean being the forward spot price satisfies Heston's solution. Thus, we show that any member of this class could be used for the direct risk neutral valuation of option prices under Heston's stochastic volatility model. In fact, we also show that any RND with mean being the forward spot price that satisfies Heston's option valuation solution must also be a member of the scale family of distributions in that mean. As particular examples, we show that under a certain re-parametrization, the one-parameter versions of the log-normal (i.e., Black–Scholes), gamma, and Weibull distributions, along with their respective inverses, are all members of this class and thus, provide explicit RNDs for direct option pricing under Heston's SV model. We demonstrate the applicability and suitability of these explicit RNDs via exact calculations and Monte Carlo simulations, using already published index data and a calibrated Heston's model (S&P500, ODAX), as well as an illustration based on recent option market data (AMD).

Keywords: Heston's model; option pricing; risk neutral valuation; calibration; negatively skewed distribution

MSC: 91G20; 60H35; 62E15

JEL Classification: G10; G13



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1. Introduction

Heston's stochastic volatility (SV) model for option valuation in [1] is now widely accepted by both academics and practitioners. It prescribes, under a risk neutral probability measure $\mathbb{Q}$, for example, the dynamics of a spot's (stock, index) price process $S = \{S_t, t \geq 0\}$ in relation to a corresponding though unobservable (and hence, untradable) volatility process $V = \{V_t, t \geq 0\}$ via a system of stochastic differential equations. This system is given by

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t} S_t dW_{1,t} \\ dV_t &= \kappa(\theta - V_t) + \eta \sqrt{V_t} dW_{2,t}, \end{aligned} \quad (1)$$

where $W_1 = \{W_{1,t}, t \geq 0\}$ and $W_2 = \{W_{2,t}, t \geq 0\}$ are two Brownian motion processes under $\mathbb{Q}$ with $d(W_1 W_2) = \rho dt$ for some $\rho \in [-1, 1]$. Here, r is the prevailing risk-free interest rate, θ and η^2 are the long-run average and variance of the volatility V , and κ is the mean reversion rate of V .

The quest to incorporate a non-constant volatility into the option valuation model when modeling the price of a European call option has arisen in the literature (e.g., [2] or [3])

ever since the seminal works of Black and Scholes in [4] and of Merton in [5] (abbreviated herein as BSM). The BSM model assumes that a spot's price evolves with the non-random and constant volatility of the spot's returns σ as a geometric Brownian motion:

$$dS_t = rS_t dt + \sigma S_t dW_{1,t}. \quad (2)$$

Coupled with the ingenious argument of instantaneous portfolio hedging (along with other assumptions, such as self-financing, no-cost trading/carry, etc.) and the application of Ito's Lemma to the underlying PDE, the BSM model provides an explicit solution for the price $C(\cdot)$ of European call options. Specifically, given the *current* spot price $S_\tau = S$ and the risk-free interest rate r , the price of the corresponding call option with price-strike K and duration T can be expressed as follows:

$$C_S(K) = S \Phi(d_1) - K e^{-rt} \Phi(d_2), \quad (3)$$

where $t = T - \tau$ is the *remaining* time to expiry. Here, using the conventional notation, $\Phi(\cdot)$ and $\phi(\cdot)$ denote the standard normal *cdf* and *pdf*, respectively, and

$$d_1 := \frac{-\log(\frac{K}{S}) + (r + \frac{\sigma^2}{2})t}{\sigma\sqrt{t}} \quad \text{and} \quad d_2 := d_1 - \sigma\sqrt{t}. \quad (4)$$

Similarly to the *form* of the BSM solution in (3), Heston found in [1] that the solution to the PDE system resulting from the stochastic volatility model (1) is given by

$$C_S(K) = S P_1 - K e^{-rt} P_2, \quad (5)$$

where P_j $j = 1, 2$ are two related (under a risk neutral probability measure $\mathbb{Q}$) conditional probabilities that the option will expire in the money, which are conditional on the given current stock price $S_\tau = S$ and the current volatility $V_\tau = V_0$. However, unlike the explicit BSM solution in (3), which is explicitly given in terms of the normal (or log-normal) distribution, Heston provided in [1] (semi)-closed-form solutions to these two probabilities: P_1 and P_2 , which are given in terms of their characteristic functions (for more details, see Appendix A). Hence, the option price $C_S(K)$ in (5) is readily computable via complex numerical integration for any of the four parameters $\vartheta = (\kappa, \theta, \eta, \rho)$ in (1), in addition to S , V_0 , and r . It should be noted that different choices for ϑ lead to different *values* of $C_S(K)$ in (5) and hence, the value of $\vartheta = (\kappa, \theta, \eta, \rho)$ must first be appropriately '*calibrated*' for $C_S(K)$ to actually match the option market prices (see Section 5 for an additional discussion). The problem of '*calibrating*' or estimating the parameters of Heston's model has naturally received much attention in the literature. Various approaches have been utilized, from standard (weighted or penalized) MSE approaches (as in [6]) to fast SWIFT method approaches (as in [7]) and direct maximum likelihood approaches (as in [8]), as well as models with implied stochastic volatility approaches (as in [9]), to name a few.

On the other hand, the utility and usefulness of the risk neutral probability measure $\mathbb{Q}$ in option valuation, both in general and in determining specific solutions in (5) (or in (3)), cannot be overstated (in the 'risk neutral' world). The risk neutral probability $\mathbb{Q}$ links together option valuation and the corresponding risk neutral distribution of a spot price S_T and its stochastic dynamics that govern the model (as in (1), for example). In the case of the BSM model with a non-random volatility in (2), the underlying RND is unique and is given by the log-normal distribution. However, this is not the case with Heston's general stochastic volatility model (1). Since it involves the dynamics of two stochastic processes $(S, \mathcal{V})$, one of which (the volatility $\mathcal{V}$) is not directly observable, there are innumerable possible choices for RNDs that could satisfy the general solutions of P_1 and P_2 in (5).

However, since ample market data are readily available on the spot prices of underlying assets, as well as their market option prices, there is an extensive body of literature dedicated to the '*extraction*', '*recovery*', '*estimation*', or even '*approximation*' of RNDs from the available market option data. For comprehensive reviews of the subject, see [10–13],

which cover both parametric and non-parametric approaches. With parametric approaches in particular, the parameters of *assumed* distributions are estimated by various means (maximum likelihood, method of moments, least squares, etc.) so as to approximate available option data or implied volatilities (c.f., [14]). These types of *assumed* multiparameter distributions include mixtures of log-normal ([12,15]), generalized gamma ([12]), generalized extreme value ([11]), gamma, and Weibull distributions ([16]), among others. While empirical considerations have often led to the suggestion that these parametric distributions are possible RNDs, the motivations for these considerations have not included any full theoretical justifications nor direct links to the governing pricing model and its dynamics as given in (1), as was the case in the BSM model, which directly (and uniquely) linked the log-normal distribution and the price dynamics of the model in (2). Accordingly, the important question of "what class of RNDs, if any, satisfies Heston's SV model and its solution (5) for option valuation?" has remained a long-standing 'open problem'.

In Theorem 2 below, we present the main result of our paper, which establishes the direct theoretical link, through Heston's solution in (5), between a particular large class of RNDs and the assumed stochastic volatility model in (1) that governs the spot price dynamics. In Section 2, we provide the theoretical justification and rationale for our approach. In Section 3, we identify the class of distributions (and hence, the class of RNDs) that satisfies Heston's solution for the option price, as is given in (5). Specifically, we show that any risk neutral probability distribution that satisfies (6) and (7) with a *scale* parameter $\mu = S \cdot e^{rt}$ (i.e., the forward spot price) would also satisfy Heston's option pricing model in (5). In fact, we also show in Appendix A that RNDs that can be calculated directly from Heston's characteristic function expressions of P_1 and P_2 must be members of this class of distributions as well. Hence, our main results of Theorem 2 provide a theoretical answer to the 'open problem' of characterizing the class of Heston's RNDs for direct option valuation. Some specific examples of well-known parametric distributions that satisfy Heston's solution are provided in Section 4. As particular examples, we show that under a certain re-parametrization, the one-parameter versions of the *log-normal* (i.e., Black–Scholes), *gamma*, and *Weibull* distributions, along with their respective inverses, are all members of this special class of distributions and thus, provide explicit RNDs for direct option pricing under Heston's SV model in (1) (or (5)). We note that a two-parameter generalization of these distributions as RNDs for Heston's SV model, namely the generalized gamma distribution, has separately been discussed by the author in [17], who has also provided several applications for real market option data.

The extent of the agreement between these distributions as possible RNDs for Heston's model, the actual Heston's RNDs (calculated numerically from expression P_2 in Appendix A), and the simulated distributions of spot prices obtained under a discretized version of the model in (1) is illustrated numerically in Section 5. The applicability and suitability of these explicit RNDs are demonstrated using already published index data and a calibrated Heston's model (S&P500 data from [18] and ODAX data from [6]), as well as an illustration based on recent market option data (AMD). In Appendix A, we present the expressions for Heston's characteristic functions and discuss some of the immediate properties leading to our main results, as stated in Theorem 2.

2. Theoretical Rationale

As has been established in [19], the risk neutral equilibrium requires that for $T > \tau$ (with $t = T - \tau$),

$$\begin{aligned}\mathbb{E}(S_T|S_\tau) &= \int S_T d\mathbb{Q}(S_T) \\ &= \int S_T \cdot q(S_T) dS_T = S_\tau e^{rt}\end{aligned}\tag{6}$$

and that (in the case of a European call option), $C_{S_\tau}(K)$ must also satisfy

$$\begin{aligned}
C_{S_\tau}(K) &= e^{-rt} \mathbb{E}((S_T - K)^+ | S_\tau) \\
&= e^{-rt} \int (S_T - K)^+ d\mathbb{Q}(S_T) \\
&= e^{-rt} \int_K^\infty (S_T - K) q(S_T) dS_T,
\end{aligned} \tag{7}$$

where, for any $x \in \mathbb{R}$, $x^+ := \max(x, 0)$. Here, $q(\cdot)$ is the risk neutral density (RND) under $\mathbb{Q}$, which is reflective of the conditional distribution of the spot price S_T at time T given the spot price S_τ at time $\tau < T$, whose expected value is the future value of the spot price (as in (6)).

By expanding the last term in (7) with $S_\tau := S$, we obtain that:

$$C_S(K) = e^{-rt} \int_K^\infty S_T \cdot q(S_T) dS_T - K e^{-rt} \mathbb{Q}(S_T > K). \tag{8}$$

Clearly, by comparing (8) to Heston's solution in (5), it follows that

$$P_2 = \mathbb{Q}(S_T > K) \equiv 1 - Q(K),$$

which is the risk neutral probability of the option expiring in the money. Further, by utilizing Equation (6) as the first term of (8), it follows from (5) that:

$$P_1 \equiv \int_K^\infty \frac{S_T}{S e^{rt}} \cdot q(S_T) dS_T = \int_K^\infty \frac{S_T}{\mathbb{E}(S_T | S_\tau)} \cdot q(S_T) dS_T. \tag{9}$$

We note that since by (6), we have

$$\int_0^\infty \frac{S_T}{\mathbb{E}(S_T | S_\tau)} \cdot q(S_T) dS_T = 1,$$

the probability P_1 is also typically interpreted as the probability of the option expiring in the money, but under the so-called physical probability measure that is dominated by $\mathbb{Q}$. However, here, in the case of Heston's SV model, we consider a different interpretation of the term P_1 , which enables us to characterize the class of RND candidates that satisfies (5).

It is standard notation to denote the so-called *delta* function (or hedging fraction) as $\Delta(K)$ in option valuation, as defined by:

$$\Delta(K) := \frac{\partial C_S(K)}{\partial S}. \tag{10}$$

In Appendix A, we show (see Claim 1) that for Heston's call option price $C_S(K)$, as given in (5), we have

$$P_1 \equiv \Delta(K), \tag{11}$$

(also see [18]). Hence, under the model in (1), Heston's solution for the option price in (5) can be written in an equivalent form as:

$$C_S(K) \equiv S \cdot \Delta(K) - K e^{-rt} \cdot (1 - Q(K)). \tag{12}$$

Accordingly, any risk neutral distribution that satisfies Heston's SV model and its solution in (5) must also admit the presentation in (12) for the direct evaluation of option price $C_S(K)$. We point out in passing that this presentation (12) also trivially applies to the BSM option price in (3) since in that case, $\Phi(d_1) \equiv \Delta(K)$ and $\Phi(d_2)$ are the probability that the option will expire in the money (as calculated under the log-normal distribution).

3. The Scale Parameter Class of Heston's RND

In this section, we identify the class of distributions (and, therefore, the class of possible RNDs) that admit the presentation in (12) for the price of a European call option. Specifically,

we show that any RND candidate that satisfies (6) and (7) with a *scale* parameter $\mu = S \cdot e^{rt}$ will admit the presentation in (12) and hence, in light of the results in (11), will equivalently satisfy Heston's option pricing model in (5). While the development given below could be seen as straightforward (and perhaps even trivial), it is instrumental for the theoretical characterization of Heston's RNDs as a *scale* family of distributions.

To that end and to simplify the presentation, we consider a continuous positive random variable X with the mean $\mu > 0$ (with respect to the underlying probability measure $\mathbb{Q}$). We denote the *cdf* and *pdf* of X as $Q_\mu(\cdot)$ and $q_\mu(\cdot)$, respectively, to emphasize their dependency on μ as a parameter. Similarly, for a given $\mu > 0$, we denote the expectation of X (or functions thereof) under Q_μ as $E_\mu(\cdot)$, so that

$$\mu := E_\mu(X) = \int_0^\infty x q_\mu(x) dx \equiv \int_0^\infty (1 - Q_\mu(x)) dx.$$

Similarly to (7), we define the following for each $s \geq 0$:

$$c_\mu(s) := E_\mu[(X - s)^+].$$

Clearly, we have $c_\mu(0) = \mu$. Note that $c_\mu(\cdot)$ is merely the *undiscounted* version of $C_S(\cdot)$ in (7), so that with $\mu = S \cdot e^{rt}$, as in (6), we have $c_\mu(K) \equiv e^{rt} \cdot C_S(K)$.

It is straightforward to see that, as in (8),

$$c_\mu(s) = \int_s^\infty (x - s) q_\mu(x) dx = \int_s^\infty x q_\mu(x) dx - s(1 - Q_\mu(s)), \quad (13)$$

or equivalently

$$c_\mu(s) \equiv \int_s^\infty (1 - Q_\mu(x)) dx. \quad (14)$$

Hence, it follows immediately from (14) that for each $s > 0$,

$$c'_\mu(s) := \frac{\partial c_\mu(s)}{\partial s} = -(1 - Q_\mu(s)). \quad (15)$$

As we proceed to explore more of the basic properties of the function $c_\mu(\cdot)$, we add the simple assumption that μ is a scale parameter of the underlying distribution of X .

Assumption A. We assume that $\mathcal{Q} := \{Q_\mu, \mu > 0\}$ is a scale family of distributions (under $\mathbb{Q}$), so that for any given $\mu > 0$,

$$Q_\mu(x) \equiv Q_1(x/\mu) \quad \text{and} \quad q_\mu(x) \equiv \frac{1}{\mu} q_1(x/\mu), \quad \forall x > 0,$$

for some *cdf* $Q_1(\cdot)$ values with a *pdf* $q_1(\cdot)$ satisfying $\int_0^\infty x q_1(x) dx = 1$ and $\int_0^\infty x^2 q_1(x) dx = 1 + v^2 < \infty$, where v^2 is an exogenous parameter to be specified later.

In Lemma 1, we establish the linear homogeneity of $c_\mu(s)$ under Assumption A and in Lemma 2, we provide the implied re-scaling property of this function and the consequential specific derived form of $c_\mu(s)$, as presented in Theorem 1. For the linear homogeneity property of European options, in general, see Merton's Theorems 6 and 9 in [5] or Theorem 2.3 in [20].

Lemma 1. Suppose that $Q_\mu \in \mathcal{Q}$ and thus, it satisfies the conditions of Assumption A. Then, the function $c_\mu(s)$, as defined in (13) or (14), is homogeneous of degree one in s and μ , i.e., for $s' = \alpha s$ and $\mu' = \alpha \mu$ with $\alpha > 0$, we have $c_{\mu'}(s') \equiv \alpha c_\mu(s)$.

Proof. With a simple change in variable, it follows immediately from Assumption A and (14) that with $s' = \alpha s$, $\mu' = \alpha \mu$, and $\alpha > 0$, we have

$$c_{\mu'}(s') = \int_{s'}^{\infty} (1 - Q_{\mu'}(x)) dx \equiv \alpha \int_{s'/\alpha}^{\infty} (1 - Q_{\mu}(u)) du = \alpha c_{\mu}(s).$$

□

Now, by applying the results of Lemma 1 to $\alpha = 1/\mu$, we immediately obtain the following useful result:

Lemma 2. Suppose that $Q_{\mu} \in \mathcal{Q}$ and thus, it satisfies the conditions of Assumption A. Then, it holds that

$$c_{\mu}(s) = \mu c_1(s/\mu), \quad (16)$$

for any $s > 0$, where c_1 is as defined in (14) but with respect to Q_1 :

$$c_1(s) = \int_s^{\infty} (1 - Q_1(u)) du \quad \text{with} \quad c_1'(s) = -(1 - Q_1(s)). \quad (17)$$

It should be clear from these results that the function $c_{\mu}(s)$ can be re-scaled or 'standardized' so that $c_{\mu}(s)/\mu$ is independent from μ . In particular, when $s = b\mu$ for some $b > 0$, then by (16), $c_{\mu}(b\mu) = \mu c_1(b)$.

Next, as in (10), we define the 'Delta' function corresponding to the function $c_{\mu}(s)$ in (13) or (14) as $\Delta_{\mu}(s) := \partial c_{\mu}(s)/\partial \mu$. In the next theorem, we show that under Assumption A, $\Delta_{\mu}(\cdot)$ can be expressed in terms of the truncated mean of X and the consequential representation of $c_{\mu}(\cdot)$.

Theorem 1. Suppose that $Q_{\mu} \in \mathcal{Q}$ and thus, it satisfies the conditions of Assumption A. Then, for each $s > 0$,

$$\Delta_{\mu}(s) = \frac{1}{\mu} \int_s^{\infty} x q_{\mu}(x) dx. \quad (18)$$

Further, $\Delta_{\mu}(s) \equiv \Delta_1(s/\mu)$, where $\Delta_1(s) := \int_s^{\infty} u q_1(u) du \leq 1$ for any $s > 0$. Hence, $c_{\mu}(s)$ in (13) can be written as

$$c_{\mu}(s) = \mu \Delta_{\mu}(s) - s(1 - Q_{\mu}(s)) \quad (19)$$

Proof. To prove (18), note that by Lemma 2, (13), and (17),

$$\begin{aligned} \Delta_{\mu}(s) &= \frac{\partial}{\partial \mu} [\mu c_1(s/\mu)] = c_1(s/\mu) - \frac{s}{\mu} c_1'(s/\mu) = \\ &= \int_{s/\mu}^{\infty} u q_1(u) du \equiv \frac{1}{\mu} \int_s^{\infty} x q_{\mu}(x) dx. \end{aligned} \quad (20)$$

The second part follows immediately from the first part and Assumption A, noting that $\Delta_1(s) \leq \int_0^{\infty} u q_1(u) du = 1$. Finally, since by (18), $\int_s^{\infty} x q_{\mu}(x) dx = \mu \cdot \Delta_{\mu}(s)$, the main result in (19) follows directly from (13). □

An immediate conclusion of Theorem 1 is that when Q_{μ} is a member of the scale family of the distribution $\mathcal{Q}$ (by Assumption A), then the $c_{\mu}(s)$ functions can easily be evaluated by first calculating the values of $c_1(\cdot)$ for the ratio $b = s/\mu$. Specifically,

$$c_{\mu}(s) = \mu c_1(s/\mu) \equiv \mu \Delta_1(s/\mu) - s(1 - Q_1(s/\mu)) \quad (21)$$

The results of the theorem, either as given in (19) or (21), can be used directly for the risk neutral valuation of a European call option with a strike K and a current spot price $S_{\tau} \equiv S$, providing the expression for $C_S(K)$ is as given in (12), i.e., when $Q \in \mathcal{Q}$, then with $\mu = S e^{rt}$ applied to (21), we have

$$\begin{aligned} C_S(K) &:= e^{-rt} \mathbb{E}((X - K)^+ | S) = e^{-rt} c_{\mu}(K) \\ &= S \Delta_1(K/\mu) - K e^{-rt} (1 - Q_1(K/\mu)). \end{aligned} \quad (22)$$

We summarize these findings in the following Corollary.

Corollary 1. *For any risk neutral distribution Q_μ that also satisfies the conditions of Assumption A, in addition to (6) and (7), with $\mu = S e^{rt}$, so that $Q_\mu \in \mathcal{Q}$ (for some $Q_1(\cdot)$), we have (as in (12)) that*

$$C_S(K) = S \Delta_\mu(K) - K e^{-rt} (1 - Q_\mu(K)),$$

where $\Delta_\mu(K) := \Delta_1(K/\mu)$ and $Q_\mu(K) := Q_1(K/\mu)$. Hence, Q_μ also satisfies Heston's option pricing model (and solution), as given in (5).

Remark 1. *In the case of a risk neutral evaluation of an option including a dividend with a rate ℓ , then $\mathbb{E}(S_T | S) = S e^{(r-\ell)t}$ in (6). In which case, by applying $\mu = S e^{(r-\ell)t}$ to (21), we obtain*

$$C_S(K) = e^{-rt} c_\mu(K) = S e^{-\ell t} \Delta_\mu(K) - K e^{-rt} (1 - Q_\mu(K)).$$

It should be clear from (22) that since the probability distribution Q_μ is assumed to be a member of a scale family $\mathcal{Q}$, its values only depend on K and S through the ratio K/S . In Appendix A, we assert in Proposition A1 that any risk neutral probability distribution Q_μ that satisfies the solution in (5) for Heston's option pricing model must also be a member of a scale family of distributions with a scale parameter $\mu = S e^{rt}$ (or $\mu = S e^{(r-\ell)t}$, in the case of a dividend yielding spot). This assertion follows directly from the specific form of Heston's RNDs established in Appendix A, which are given in terms of the characteristic functions corresponding to the term P_2 (see (A5) and the subsequent comments). Hence, when combined, Corollary 1 and Proposition A1 provide the main results of the paper, which are summarized in the following theorem.

Theorem 2. *Let $Q_\mu(\cdot)$ be any risk neutral distribution that satisfies (6) and (7) with a corresponding RND $q_\mu(\cdot)$ and with $\mu := S e^{rt}$ being the forward price. Then, $Q_\mu(\cdot)$ satisfies Heston's option pricing solution in (5) (and equivalently in (12)) if, and only if, $Q_\mu(\cdot)$ is a member of a scale family of distributions $\mathcal{Q}$ with a scale parameter μ .*

Clearly, Theorem 2 asserts that any RND for Heston's stochastic volatility model must be a member of the class of scale parameter distributions that satisfies Assumption A, with the mean being the forward spot price. Indeed, any member of this class of scale parameter distributions could serve as a possible RND for the direct and risk neutral calculation of an option price, as in (7) or Corollary 1. In the next section, we identify and list several one-parameter versions of some well-known distributions that could serve as RNDs under Heston's SV model.

4. Examples of Explicit RNDs for Heston's Model

In view of Theorem 2, the quest for finding appropriate RNDs for Heston's SV model for particular parametrizations of $\vartheta = (\kappa, \theta, \eta, \rho)$ must be focused only on the members of a scale family of distributions with a scale parameter $\mu = S e^{rt}$. Accordingly, in this section, we provide five particular examples of well-known distributions that satisfy the conditions of Assumption A and hence, admit, as per Corollary 1, the presentation (12) for Heston's option pricing model in (5). These well-known distributions, namely, the *log-normal*, *gamma*, *inverse Gaussian*, *Weibull*, and *inverse Weibull* distributions, are re-parametrized under Assumption A to a standardized, one-parameter versions, denoted as $Q_1(\cdot | \nu)$, with the mean 1 and a second moment that depends on a single parameter $\nu > 0$ (in fact, we later take $\nu \equiv \sigma \sqrt{t}$ for some $\sigma > 0$). Due to their relative simplicity (i.e., involving only one parameter), we view these distributions as *computationally inexpensive* RNDs, which are easier to obtain, calculate, and calibrate compared to the alternatives approaches available in the literature. We note that while the gamma and Weibull distributions were considered in [16,21] for deriving 'alternative' option pricing formulae, the motivation for

the parametrization discussed there did not consider the spot price dynamics (i.e., as in the model in (1)) but rather fitting kurtosis and skew; therefore, they are inherently different.

With these standardized distributions in hand and the corresponding explicit expressions for $Q_1(\cdot) \equiv Q_1(\cdot|\nu)$ as obtained under *Assumption A*, we utilize (21) (with $\mu \equiv 1$) to first calculate in each case the expression for

$$c_1(s) = \Delta_1(s) - s(1 - Q_1(s)),$$

which is then used to obtain the expression for the *undiscounted* option price, with $\mu > 0$:

$$c_\mu(s) \equiv \mu \times c_1(s/\mu) = \mu \times \left[\Delta_1(s/\mu) - \frac{s}{\mu} \times (1 - Q_1(s/\mu)) \right].$$

Finally, as in (22), the corresponding expression for the call option price is obtained as $C_S(K) = e^{-rt} c_\mu(K)$ (with $\mu = Se^{rt}$, $s = K$, and $\nu = \sigma\sqrt{t}$). We point out again that each of these five distributions would satisfy Heston's general solution for the valuation of European call options as RNDs, as given in (5). We begin with the log-normal distribution, which results in the classical Black–Scholes option pricing model (as given in (3) and (4)).

4.1. The Log-Normal RND

Suppose that the random variable U has the 'standard' (one-parameter) log-normal distribution with the mean $E(U) = 1$ and variance $Var(U) = e^{\nu^2} - 1$ for some $\nu > 0$, so that $W = \log(U) \sim \mathcal{N}(-\nu^2/2, \nu^2)$. Accordingly, the *pdf* of U is given by

$$q_1(u) = \frac{1}{u\nu} \times \phi\left(\frac{\log(u) + \nu^2/2}{\nu}\right), \quad u > 0,$$

and the *cdf* of U is

$$Q_1(u) = Pr(U \leq u) = \int_0^u q_1(s) ds = \Phi\left(\frac{\log(u) + \nu^2/2}{\nu}\right), \quad \forall u > 0.$$

It is straightforward to verify that when $X \equiv \mu U$ for some $\mu > 0$, then the *pdf* of X is the 'scaled' version of q_1 , namely, $q_\mu(x) = \frac{1}{\mu} q_1(x/\mu)$. So, this distribution satisfies *Assumption A*.

Next, we calculate the expression of $\Delta_1(s)$, which upon using the relation $U \equiv e^W$, becomes

$$\begin{aligned} \Delta_1(s) &:= \int_s^\infty u q_1(u) dx = \int_{\log(s)}^\infty e^w \phi\left(\frac{w + \nu^2/2}{\nu}\right) \frac{dw}{\nu} \\ &= \int_{\log(s)}^\infty \phi\left(\frac{w - \nu^2/2}{\nu}\right) \frac{dw}{\nu} = 1 - \Phi\left(\frac{\log(s) - \nu^2/2}{\nu}\right). \end{aligned}$$

Hence, for the 'standardized' model, we have that

$$\begin{aligned} c_1(s) &= \Delta_1(s) - s(1 - Q_1(s)) \equiv \\ &\left[1 - \Phi\left(\frac{\log(s) - \nu^2/2}{\nu}\right) \right] - s \left[1 - \Phi\left(\frac{\log(s) + \nu^2/2}{\nu}\right) \right]. \end{aligned}$$

Accordingly, by Lemma 2 and (21), $c_\mu(s) \equiv \mu \times c_1(s/\mu)$ and, therefore, we immediately obtain the following expression for $c_\mu(s)$:

$$\begin{aligned} c_\mu(s) &= \mu \times \left[\Delta_1(s/\mu) - \frac{s}{\mu} \times (1 - Q_1(s/\mu)) \right] \\ &= \mu \times \left[1 - \Phi\left(\frac{\log(s/\mu) - \nu^2/2}{\nu}\right) \right] - s \times \left[1 - \Phi\left(\frac{\log(s/\mu) + \nu^2/2}{\nu}\right) \right] \\ &\equiv \mu \times \Phi\left(\frac{\log(\mu/s) + \nu^2/2}{\nu}\right) - s \times \Phi\left(\frac{\log(\mu/s) - \nu^2/2}{\nu}\right), \end{aligned}$$

where the last equality utilizes the symmetry of the normal distribution. Finally, to calculate the price of a call option under the log-normal RND at a strike K when the current price of the spot is S , we utilize the above $c_\mu(s)$ expression with $\mu \equiv S e^{rt}$, $s \equiv K$, and $\nu \equiv \sigma\sqrt{t}$ to obtain $C_S(K) = e^{-rt}c_\mu(K)$, which exactly matches the Black–Scholes call option price, as given in (3) and (4).

4.2. The Gamma RND

We begin with some standard notations. We write $W \sim \mathcal{G}(\alpha, \lambda)$ to indicate that the random variable W has the gamma distribution with a scale parameter $\lambda > 0$ and a shape parameter $\alpha > 0$, in which case we write $g(\cdot; \alpha, \lambda)$ and $G(\cdot; \alpha, \lambda)$ for the corresponding *pdf* and *cdf* values of W , respectively. Recall that $E(W) = \alpha/\lambda$ and $Var(W) = \alpha/\lambda^2$. Additionally, we denote the gamma function whose incomplete version is $\Gamma(\xi, \alpha) := \int_0^\xi y^{\alpha-1} e^{-y} dy$ as $\Gamma(\alpha) := \int_0^\infty y^{\alpha-1} e^{-y} dy$, which is defined for any $\xi > 0$.

Now, suppose that a random variable U has the ‘standard’ (one-parameter) gamma distribution with the mean $E(U) = 1$ and variance $Var(U) = \nu^2$ for some $\nu > 0$, so that $U \sim \mathcal{G}(a, a)$, where we substitute $a \equiv 1/\nu^2$. Accordingly, the *pdf* of U is given by

$$q_1(u) := g(u; a, a) = \frac{a(au)^{a-1} e^{-au}}{\Gamma(a)}, \quad u > 0.$$

and the *cdf* of U is given by

$$Q_1(u) = Pr(U \leq u) := G(u; a, a) = \frac{\Gamma(au, a)}{\Gamma(a)},$$

for any $u > 0$. It is straightforward to verify that when $X \equiv \mu U$ for some $\mu > 0$, then the *pdf*, $q_\mu(\cdot)$ of X is the ‘scaled’ version of $q_1(\cdot)$ and that Assumption A holds in this case too. Next, we calculate the expression for $\Delta_1(s)$:

$$\begin{aligned} \Delta_1(s) &= \int_s^\infty u q_1(u) du = \int_s^\infty u g(u; a, a) du \\ &= \int_s^\infty \frac{a(au)^a e^{-au}}{a\Gamma(a)} du = 1 - \frac{\Gamma(as, a+1)}{\Gamma(a+1)} \equiv 1 - G(s; a+1, a). \end{aligned}$$

Accordingly, for the ‘standardized’ gamma model, we obtain that

$$\begin{aligned} c_1(s) &= \Delta_1(s) - s(1 - Q_1(s)) \equiv \\ &= \left[1 - \frac{\Gamma(as, a+1)}{\Gamma(a+1)} \right] - s \left[1 - \frac{\Gamma(as, a)}{\Gamma(a)} \right] \\ &= [1 - G(s; a+1, a)] - s[1 - G(s; a, a)]. \end{aligned}$$

Again, by Lemma 2 and (21), $c_\mu(s) \equiv \mu \times c_1(s/\mu)$ and, therefore, we immediately obtain the following expression for $c_\mu(s)$ in the case of the gamma model:

$$\begin{aligned} c_\mu(s) &= \mu \times \left[\Delta_1(s/\mu) - \frac{s}{\mu} \times (1 - Q_1(s/\mu)) \right] \\ &= \mu \times [1 - G(s/\mu; a+1, a)] - s \times [1 - G(s/\mu; a, a)]. \end{aligned} \quad (23)$$

Finally, to calculate the price of a call option under this gamma RND at a strike K when the current price of the spot is S , we utilize (23) with $\mu \equiv S e^{rt}$, $s \equiv K$, and $\nu \equiv \sigma\sqrt{t}$ (so that $a \equiv 1/\sigma^2 t$) to obtain $C_S(K) = e^{-rt}c_\mu(K)$.

4.3. The Inverse Gaussian RND

Using standard notation, we write $W \sim \mathcal{IN}(\alpha, \lambda)$ to indicate that the random variable W has the inverse Gaussian distribution with the mean $E(W) = \alpha$ and $Var(W) = \alpha^3/\lambda$. Now, suppose that a random variable U has the ‘standard’ (one-parameter) inverse Gaus-

sian distribution with the mean $E(U) = 1$ and variance $Var(U) = \nu^2$ for some $\nu > 0$, so that $U \sim \mathcal{IN}(1, 1/\nu^2)$. Accordingly, the *pdf* and *cdf* values of U are given by

$$q_1(u) = \frac{1}{\nu u^{3/2}} \times \phi\left(\frac{u-1}{\nu\sqrt{u}}\right), \quad u > 0,$$

and

$$Q_1(u) = \Phi\left(\frac{u-1}{\nu\sqrt{u}}\right) + e^{2/\nu^2} \times \Phi\left(-\frac{u+1}{\nu\sqrt{u}}\right), \quad \forall u > 0.$$

Again, we can verified that when $X \equiv \mu U$ for some $\mu > 0$, then the *pdf*, $q_\mu(\cdot)$ of X is the 'scaled' version of $q_1(\cdot)$, so *Assumption A* holds for this distribution too. In this case, the values of $\Delta_1(s) = \int_s^\infty u q_1(u) du$ must be evaluated numerically and when combined with the expression of $Q_1(s)$ given above, they provide the values of

$$c_\mu(s) = \mu \times \left[\Delta_1(s/\mu) - \frac{s}{\mu} \times (1 - Q_1(s/\mu)) \right],$$

for any $\mu > 0$. Here, again, the corresponding values of the call option $C_S(K)$ can be obtained through exactly the same lines as in the previous examples, with $\mu \equiv S e^{rt}$, $s \equiv K$, and $\nu = \sigma\sqrt{t}$.

4.4. The Weibull RND

Using standard notation, we write $W \sim \mathcal{W}(\xi, \lambda)$ to indicate that the random variable W has the Weibull distribution with *cdf* and *pdf* values in the form of

$$F_W(w) = 1 - e^{-(w/\lambda)^\xi}, \quad \text{and} \quad f_W(w) = \frac{\xi}{\lambda} \left(\frac{w}{\lambda}\right)^{\xi-1} e^{-(w/\lambda)^\xi}, \quad w > 0,$$

respectively, where $\lambda > 0$ is the scale parameter and $\xi > 0$ is the shape parameter. The mean and variance of W are given by

$$E(W) = \lambda h_1(\xi) \quad \text{and} \quad Var(W) = \lambda^2 (h_2(\xi) - h_1(\xi)^2),$$

where $h_j(\xi) := \Gamma(1 + j/\xi)$, $j = 1, \dots, 4$. Now, suppose that a random variable U has the 'standard' (one-parameter) Weibull distribution with the mean $E(U) = 1$ and variance $Var(U) = \nu^2$ for some $\nu > 0$, i.e., for a given $\nu > 0$, we let $\xi^* \equiv \xi(\nu)$ be the (unique) solution of the following equation:

$$\frac{h_2(\xi)}{h_1^2(\xi)} = 1 + \nu^2, \quad (24)$$

in which case, $h_j^* \equiv h_j(\xi^*)$, $j = 1, 2$, $\lambda^* \equiv 1/h_1^*$, and $U \sim \mathcal{W}(\xi^*, \lambda^*)$. Accordingly, the *pdf* and *cdf* values of U are given by

$$Q_1(u) = 1 - e^{-(u/\lambda^*)^{\xi^*}}, \quad \text{and} \quad q_1(u) = \frac{\xi^*}{\lambda^*} \left(\frac{u}{\lambda^*}\right)^{\xi^*-1} e^{-(u/\lambda^*)^{\xi^*}}, \quad u > 0.$$

Again, it can be easily verified that when $X \equiv \mu U$ for some $\mu > 0$, then the *pdf*, $q_\mu(\cdot)$ of X is the 'scaled' version of $q_1(\cdot)$, so *Assumption A* holds in this case too. For this RND, the values of $\Delta_1(s)$ can be obtained in closed form as

$$\Delta_1(s) = \int_s^\infty u q_1(u) du = 1 - \frac{\Gamma((s/\lambda^*)^{\xi^*}; 1 + 1/\xi^*)}{\Gamma(1 + 1/\xi^*)},$$

which, together with the expression of $Q_1(\cdot)$ given above, provide the values of

$$c_\mu(s) = \mu \times \left[\Delta_1(s/\mu) - \frac{s}{\mu} \times (1 - Q_1(s/\mu)) \right],$$

for any $\mu > 0$. Here, again, the corresponding values of the call option $C_S(K)$ can be obtained through exactly the same lines as in the previous examples, with $\mu \equiv S e^{rt}$, $s \equiv K$, and $\nu = \sigma\sqrt{t}$.

4.5. The Inverse Weibull RND

Similarly to the above example, we write $W \sim \mathcal{IW}(\xi, \alpha)$ to indicate that the random variable W has the inverse Weibull distribution with *cdf* and *pdf* values in the form of

$$F_W(w) = e^{-(\alpha/w)^\xi}, \quad \text{and} \quad f_W(w) = \frac{\xi}{\alpha} \left(\frac{\alpha}{w}\right)^{\xi+1} e^{-(\alpha/w)^\xi}, \quad w > 0, \quad (25)$$

respectively, where $\alpha > 0$ is the scale parameter and $\xi > 2$ is the shape parameter. In this case, the mean and variance of W are given by

$$E(W) = \alpha \tilde{h}_1(\xi) \quad \text{and} \quad \text{Var}(W) = \alpha^2 (\tilde{h}_2(\xi) - \tilde{h}_1^2(\xi)),$$

where $\tilde{h}_j(\xi) \equiv h_j(-\xi) = \Gamma(1 - j/\xi)$, $j = 1, \dots, 4$. Here, too, we let U have the 'standard' (one-parameter) inverse Weibull distribution with the mean $E(U) = 1$ and variance $\text{Var}(U) = \nu^2$ for some $\nu > 0$, i.e., for a given $\nu > 0$, we let $\xi^* \equiv \xi(\nu)$ be the (unique) solution of the following equation:

$$\frac{\tilde{h}_2(\xi)}{\tilde{h}_1^2(\xi)} = 1 + \nu^2, \quad (26)$$

in which case, $U \sim \mathcal{IW}(\xi^*, \alpha^*)$ with $\alpha^* = 1/\tilde{h}_1(\xi^*)$. Accordingly, the *pdf*, $q_1(u)$ and *cdf*, $Q_1(u)$ of U are as given in (25) but with ξ^* and α^* . Hence, we can proceed along exactly the same lines as the previous examples to calculate $c_1(s)$ and $c_\mu(s)$ and hence, $C_S(K) = e^{-rt} c_\mu(K)$ (with $\mu = S e^{rt}$, $s = K$, and $\nu = \sigma\sqrt{t}$).

4.6. On Skewness and Kurtosis

As already seen, the distributions in each of the five examples satisfy the conditions of *Assumption A* and hence, by Corollary 1, could potential serve as RNDs for Heston's SV model in (1). These distributions are defined by a single parameter, namely $\nu \equiv \sigma\sqrt{t}$, which affects their features, such as *kurtosis* and *skewness*, and hence, their suitability as RNDs for each particular scenario in the SV model (1), as determined by the structural model parameter $\vartheta = (\kappa, \theta, \eta, \rho)$ (more on this point in the next section). However, for the sake of completeness and for future reference, we provide the expressions for the *kurtosis* and *skewness* of these five distributions in Table 1.

Table 1. The *skewness* and excess *kurtosis* measures of the RND examples (3.1–3.5) as functions of the single parameter $\nu \equiv \sigma\sqrt{t}$.

Distribution	$E(U)$	$\text{Var}(U)$	<i>Skew</i>	<i>Exc.Kurtosis</i>
$\mathcal{G}(1/\nu^2, 1/\nu^2)$	1	ν^2	2ν	$6\nu^2$
$\mathcal{IN}(1, 1/\nu^2)$	1	ν^2	3ν	$15\nu^2$
$\mathcal{W}(\xi, 1/g_1(\xi))^*$	1	ν^2	$\gamma_s(\xi)$	$\gamma_k(\xi) - 3$
$\mathcal{IW}(\xi, 1/h_1(\xi))^{**}$	1	ν^2	$\gamma_s(-\xi)$	$\gamma_k(-\xi) - 3$
$\mathcal{LN}(-\nu^2/2, \nu^2)$	1	$e^{\nu^2} - 1$	$(e^{\nu^2} + 2)\sqrt{e^{\nu^2} - 1}$	$e^{4\nu^2} + 2e^{3\nu^2} + 3e^{2\nu^2} - 6$

* $\xi \equiv \xi(\nu)$ solves Equation (24); ** $\xi \equiv \xi(\nu)$ solves Equation (26). It is assumed that ν is such that $\xi(\nu) > 4$ for these expressions to be valid, in particular, with $h_j(\xi) = \Gamma(1 + j/\xi)$, $j = 1, \dots, 4$, $\gamma_s(\xi) = \frac{h_3(\xi) - 3h_2(\xi)h_1(\xi) + 2h_1^3(\xi)}{[h_2(\xi) - h_1^2(\xi)]^{3/2}}$,

and $\gamma_k(\xi) = \frac{h_4(\xi) - 4h_3(\xi)h_1(\xi) + 6h_2(\xi)h_1^2(\xi) - 3h_1^4(\xi)}{[h_2(\xi) - h_1^2(\xi)]^2}$.

It is interesting to note that in the relevant parametric domain, all but the Weibull example have positive *skewness* measures. It can be numerically verified that in the Weibull case, $\gamma_s(\xi(\nu))$ changes its sign and is negative once $\nu < 0.3083511$ and that $\gamma_k(\xi(\nu)) < 3$ for $0.2007844 < \nu < 0.4698801$ and is a largely leptokurtic distribution for $\nu > 0.4698801$. Hence, the Weibull distribution could be particularly useful when the implied RND is expected to be negatively skewed, such as when the spot is an index (more on this point in the next section).

5. Comparisons of Heston's RNDs

Having introduced several examples of distributions that could serve as possible RNDs for Heston's option valuation (5) under the stochastic volatility model (1) in the previous section, we dedicate this section to the illustration of their applicability and their relative comparison. In Appendix A, we provide the closed form expressions for Heston's P_1 and P_2 , which are given in terms of their characteristic functions (see [1]). These expressions enable us to compute Heston's call price $C_S(K)$ for given $S_\tau = S$, $V_\tau = V_0$, and r values and for each choice of $\vartheta = (\kappa, \theta, \eta, \rho)$, as in (5), as well as calculate RNDs that are derived from Heston's characteristic functions of P_2 (see (A5) in Appendix A for details on $\tilde{p}_2(\cdot)$, the exact RND (under $\mathbb{Q}$) for Heston's model). The features of this distribution, such as *kurtosis* and *skewness*, are largely determined by η and ρ , respectively (see [18]) and could serve as guides for matching particular proposed RNDs from among our five examples (also see Table 1), all corresponding to a particular 'scenario', as determined by $\vartheta = (\kappa, \theta, \eta, \rho)$. For instance, in cases which admit a RND with a distinct negative skew, the *Weibull* distribution could be considered, whereas in cases with a distinct positive skew, the *inverse Weibull* or one of the other distributions discussed in Section 4 could be considered.

Additionally, utilizing a discretized version of Heston's stochastic volatility process (1), we can simulate observations on (S_T, V_T) to obtain simulated renditions of the marginal distributions of S_t and V_t (which are conditional on S and V_0 , with $t = T - \tau$). In light of the scaling property of the RNDs, we present the results in terms of the rescaled spot prices $S_t^* = S_t/\mu$ whenever convenient, where $\mu = Se^{rt}$ (see Corollary 1). In the simulations, we employ either the (reflective version of) bMilestein's discretization scheme (see [22]) or Alfonsi's implicit discretization scheme (see [23]), depending on whether the so-called Feller condition (i.e. [24]), $\zeta := 2\kappa\theta/\eta^2 > 1$ holds or not (see [25] for a discussion). We note that ζ is intimately related to the conditional distribution of V_t that is implied by the SV model in (1) (see Proposition 2 of [26] for details). In fact, it can be verified from [27] that the conditional distribution of V_t given V_0 can be expressed under the parametrization $a = e^{-\kappa t}$ and $b = 2\zeta a/(\theta(1-a))$ as

$$V_t^* := \frac{bV_t}{a} \sim \chi^2[2\zeta, bV_0], \quad (27)$$

which is the non-central chi-squared distribution with 2ζ degrees of freedom and a parameter of non-centrality of bV_0 .

To illustrate this relation, we draw $M = 10,000$ simulated pairs (S_t^*, V_t^*) with a standardized spot price and volatility, according to the SV model in (1) as described above, with (arbitrarily chosen) $\kappa = 1.150$, $\theta = 0.035$, $\eta = 0.390$, and $\rho = -0.640$ and with $S_0 = 100$, $V_0 = 0.05$, $r = 0.01$, and $t = 56/365$. Their joint distribution is presented in Figure 1a, where we have superimposed the matching 16%, 50%, 68%, 95%, and 99.5% contour lines. In Figure 1b, we present a histogram of the simulated marginal distribution of the spot price S^* , along with the theoretical Heston's RND calculated using (A5). In Figure 1c, we display a histogram of the simulated marginal distribution of the standardized volatility V^* , along with the theoretical chi-squared density curve, according to (27).

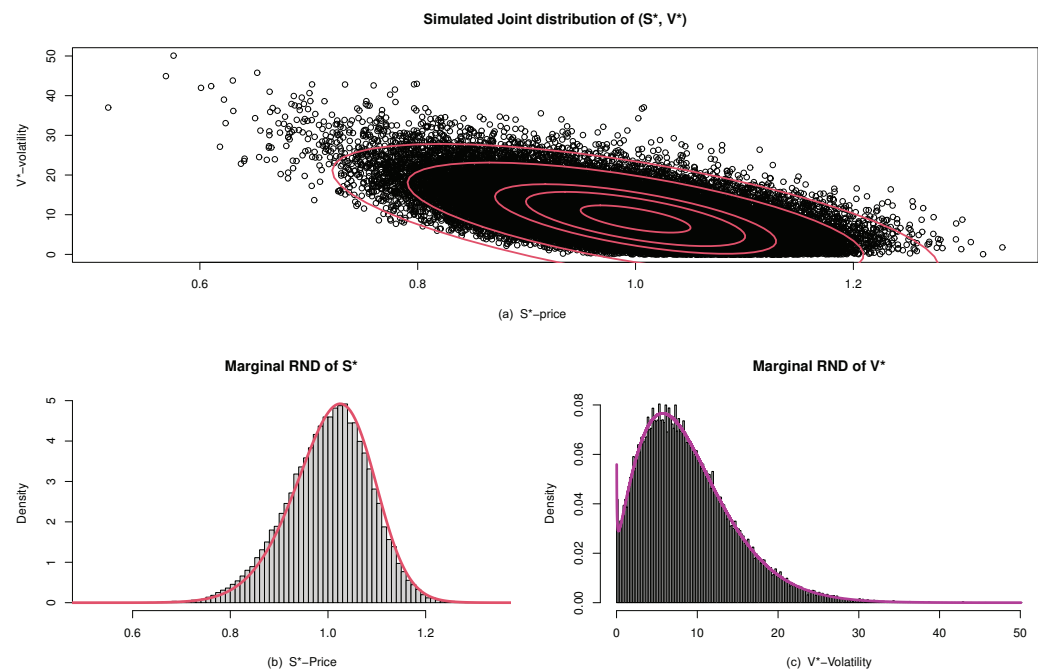


Figure 1. Illustrations of the simulated joint (S^*, V^*) distribution, along with marginal histograms of the simulated S^* and V^* values and their theoretical marginal distributions, according to (A5) for S^* and (27) for V^* .

The same approach is utilized in the following examples, which are intended to illustrate the extent of the agreement between our proposed class of RNDs in the various cases and Heston's semi-closed form RNDs for S^* , which can be calculated using (A5) under the 'calibrated' values of $\vartheta = (\kappa, \theta, \eta, \rho)$. In all cases, we also include a comparison to the Monte Carlo distribution of S^* , which is generated using the corresponding discretized version of model (1).

In the first two examples, we use values that are already calibrated to market data for the structural parameters $\vartheta = (\kappa, \theta, \eta, \rho)$, which can be found in [18] (on the S&P500) and in [6] (on the ODAX). These two examples, involving market data on traded indexes are used to illustrate the applicability of the Weibull distribution to situations in which the RND is negatively skewed and largely leptokurtic. Other similar illustrations using calibrated parameter values, such as those in [28] (on the EURO STOXX 50), are also available but not presented here due to limited space. To allow for the additional comparisons to be as realistic as possible, the next example is based on current (as of 31 December 2020) AMD market option data. This example serves to illustrate the applicability of the other RND candidates discussed in Section 4 to situations exhibiting mildly positive skewness.

Example (S&P500). Bakshi, Cao, and Chen presented in [18] an extensive market data study comparing several competing stochastic volatility models, including Heston's model. The data used in their study covered options and spot prices for the S&P500 Index from 1 June 1988 to 31 May 1991. From their Table III (see page 2008 in [18]), we find that in addition to $r = 0.02$, the 'All Option' estimated (or implied) structural parameter corresponding Heston's SV model is

$$\hat{\vartheta} = (1.15, (0.04/1.15), 0.39, -0.64).$$

In this case, $\hat{\zeta} = 0.526 < 1$; hence, for a short contract duration with $t = 56/365 = 0.153$ year, we used Milstein's (reflective) scheme to obtain a Monte Carlo sample of $M = 10,000$ simulated pairs (S^*, V) with a standardized spot price and volatility, according to the SV model in (1). The results of their joint distribution is presented in Figure 2a. In Figure 2b, we present a histogram of the simulated marginal distribution of the spot price S^* . The mean and standard deviation of these M simulated spot price values are $\bar{S}^* = 0.999462$ and $\hat{\sigma}\sqrt{t} = 0.07213028$. We also include in the figure the curve of the

implied (by $\hat{\vartheta}$) Heston's RND, as computed directly using (A5). As is expected in the case of risk neutrality modeling the spot prices of an index, the implied RND is **negatively skewed** ($sk = -0.5018587$), which suggests that it is comparable to the Weibull distribution discussed in Section 4.4. To that end and since we do not have the actual option data used by [18], we simply matched ν to the 'observed' value of $\hat{\sigma}\sqrt{t}$ and use this to obtain the numerical solution of Equation (24) as $\hat{\xi} = 17.40468$, where $h_1(\hat{\xi}) = 0.9699386$. For comparison, we also include the plot of the $\mathcal{W}(\hat{\xi}, 1/h_1(\hat{\xi}))$ RND in Figure 2b. As can be seen, the two RND curves are almost indistinguishable.

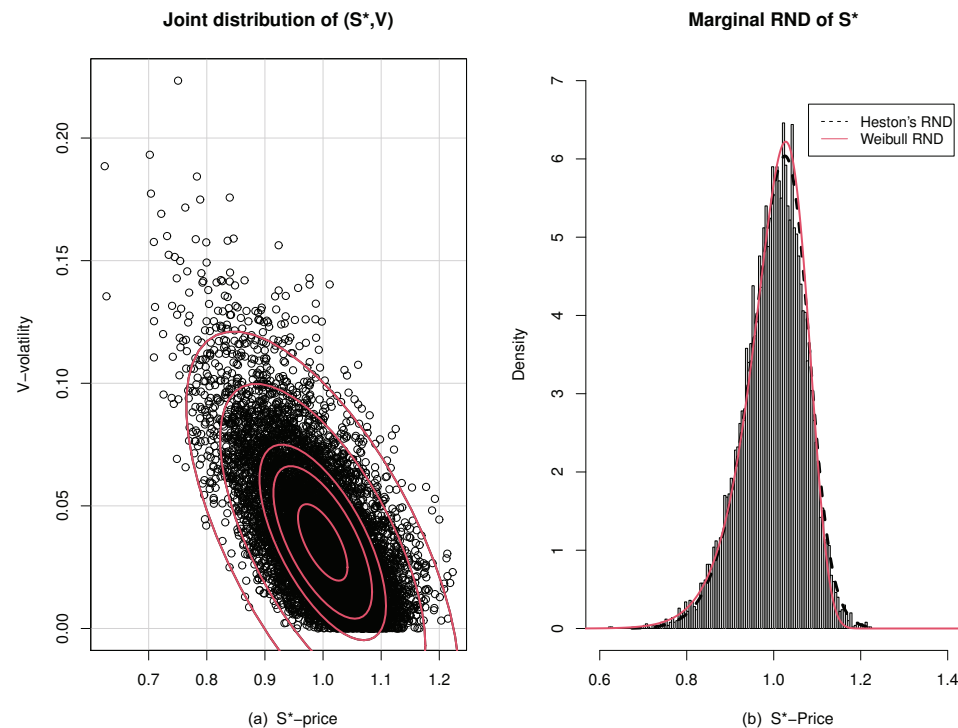


Figure 2. The simulated joint (S^*, V) distribution and the Heston's and Weibull RNDs for the S&P500 data, based on the calibrated parameter $\hat{\vartheta} = (1.15, 0.0347826, 0.39, -0.64)$ provided in [18].

To further illustrate the applicability of the proposed RNDs to situations with distinctly **positive skewness**, we again consider the calibrated parametrization of [18] that we used for Figure 2 but with a *hypothetically* positive correlation, so that $\hat{\vartheta} = (1.15, (0.04/1.15), 0.39, 0.64)$. The simulated Monte Carlo distributions (both joint and marginal) are presented in Figure 3, exhibiting the distinctly **positive skewness** of the Heston's RND (calculated from (A5)). This suggests that it is comparable to the inverse Weibull distribution discussed in Section 4.5. The mean and standard deviation of these M simulated spot price values are $\bar{S}^* = 0.9980681$ and $\hat{\sigma}\sqrt{t} = 0.07379416$. Again, the value of ν is matched to $\hat{\sigma}\sqrt{t}$ to obtain the solution of Equation (26) as $\hat{\xi} = 18.16455$, where $\tilde{h}_1(\hat{\xi}) = 1.034936$. Accordingly, we include the plot of the $\mathcal{IW}(\hat{\xi}, 1/\tilde{h}_1(\hat{\xi}))$ RND to Figure 3b for comparison, illustrating the extent of the agreement between the Heston's (implied) RND and the inverse Weibull distribution in this case (with a distinctly positive skew).

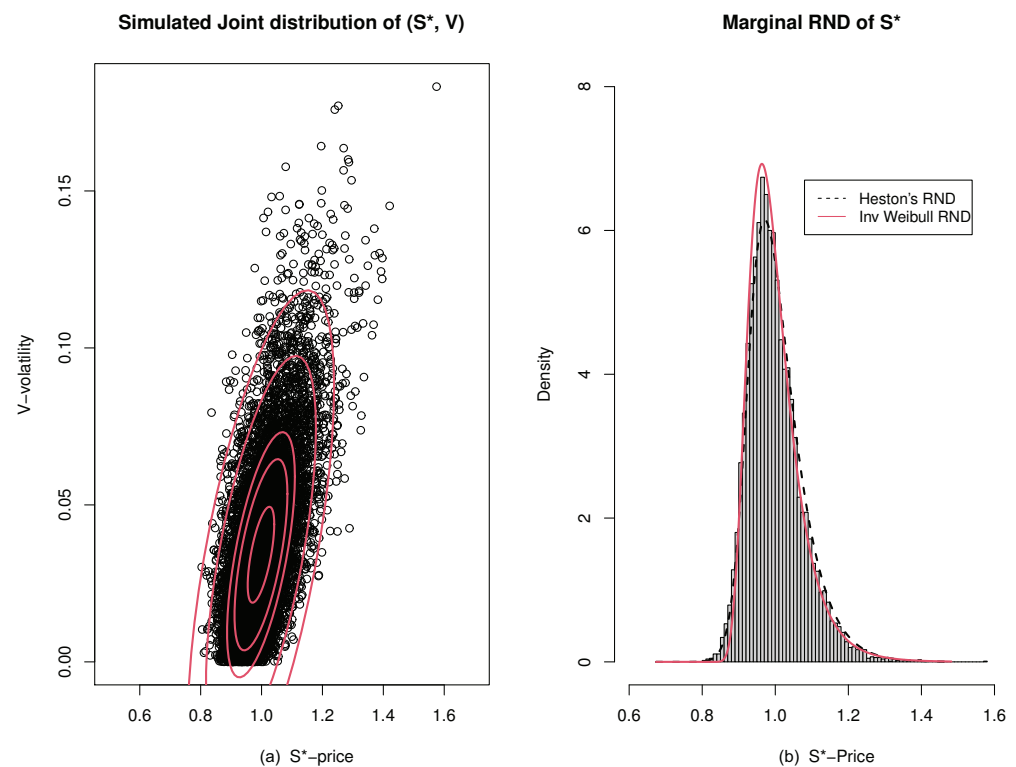


Figure 3. Illustrations of the inverse Weibull *pdf* for the S&P500 data from [18] with a calibrated parameter but with hypothetically positive correlation ($\rho = 0.64$), resulting with a positively skewed RND.

Example (ODAX): The authors of [6] studied various optimization techniques for calibrating and simulating Heston's model. To demonstrate their results, they used the ODAX option index with five blended maturities of 3 and 6 months and with 107 strikes, which were recorded on 19 March 2013. The calibrated results of the structural parameter $\theta = (\kappa, \theta, \eta, \rho)$ are provided in their Table 4 (see page 698 in [6]):

$$\hat{\theta} = (1.22136, 0.06442, 0.55993, -0.66255).$$

with $r = 0.00207$ and a 'current' spot price of $S = 7962.31$ and $V_0 = 0.02497$. Under this parametrization, we simulate a total of $M = 10,000$ pairs of (S^*, V) with $t = 64/365$ to obtain the renditions of their joint distribution from the discretized process, as well as the marginal distribution of S^* . These are presented in Figure 4. The mean and standard deviation of these M simulated spot price values are $\bar{S}^* = 0.9989159$ and $\hat{\sigma}\sqrt{t} = 0.0692782$. In the figure, the Heston's RND calculated under this parametrization from the characteristic function given in Appendix A (i.e., using (A5)) is also superimposed. Again, as is expected for this index, the implied RND is **negatively skewed** ($sk = -0.814462$), which suggests that it is comparable to the Weibull distribution. In this case too, the value of ν is matched to $\hat{\sigma}\sqrt{t}$ to obtain the solution of Equation (26) as $\hat{\xi} = 19.90341$, where $\tilde{h}_1(\hat{\xi}) = 0.9733867$. The graph of the Weibull distribution $\mathcal{W}(\hat{\xi}, 1/h_1(\hat{\xi}))$ is also displayed in the figure and indicates the excellent agreement between the calculated Heston's RNDs.

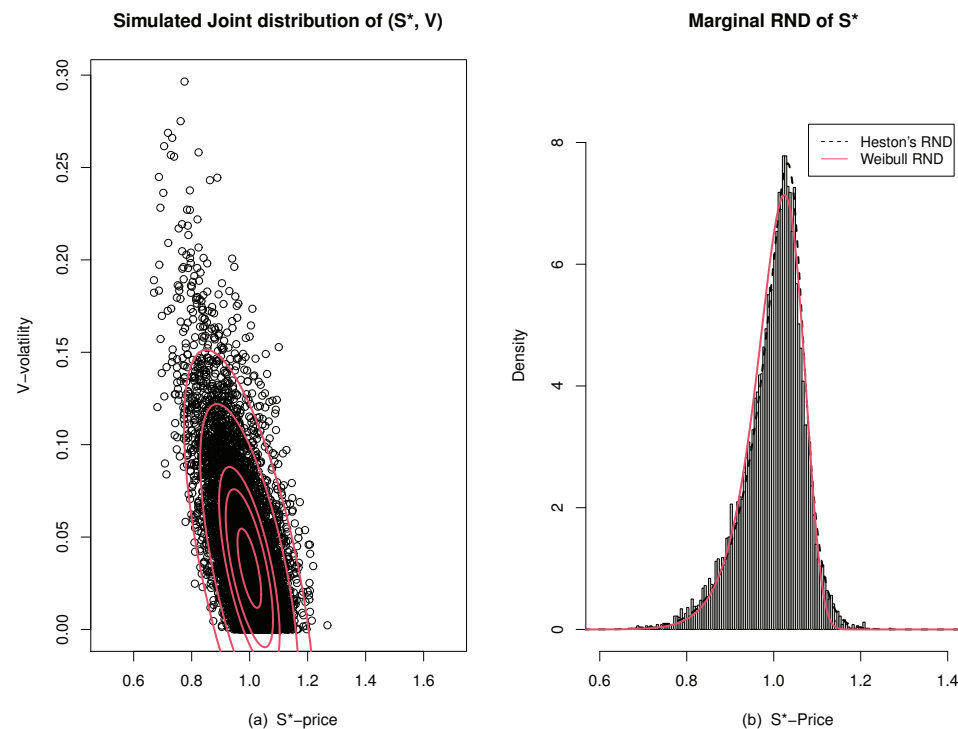


Figure 4. The simulated joint (S^*, V) distribution and the conditional Heston's and Weibull RNDs for the ODAX data, based on the calibrated parameter $\hat{\vartheta} = (1.22136, 0.06442, 0.55993, -0.66255)$ provided by [6].

Example (AMD): This example is based on recent real option data we retrieved from *Yahoo Finance* as of the closing of trading on 31 December 2020. On that day, the closing price of this stock was 91.71 and it paid no dividend, so $q = 0$ was added to the prevailing (risk-free) interest rate of $r = 0.0016$. We chose this stock (AMD, Advanced Micro Devices Inc.), which is a member of the technology sector, since it exhibits more directional risk to the upside and hence, would produce potentially positively skewed RNDs.

From the available option series, we selected the 19 February 2021 expiry due to the relatively short contract with $t = 47/365$, $N = 39$ strikes, and $K_1, \dots, K_{39}$ with corresponding available call option (market) prices $C_1, \dots, C_{39}$ (we actually recorded the option prices as the averages between the bids and asks). As the standard measure of the *goodness-of-fit* between the model-calculated option price $C^{Model}(K_i)$ and the option market price C_i , we use the *mean squared error* (MSE):

$$MSE(Model) = \frac{1}{N} \sum_{i=1}^N (C^{Model}(K_i) - C_i)^2$$

To calibrate Heston's SV model, we use the `optim()` function of R to minimize $MSE(Heston)$ over the model's parameter $\vartheta = (\kappa, \theta, \eta, \rho)$ with the initial values of $(2, 0.5, 0.6, 0)$ and $V_0 = 0.25$. The results of the calibrated values are

$$\hat{\vartheta} = (1.38164142, 1.06637168, 1.72832698, 0.07768964).$$

The calibrated parameter $\hat{\vartheta}$ is then used to calculate the option prices according to Heston's SV model in (5) using Heston's characteristic function (i.e., (A5)). These values are displayed in Table 2, along with the actual observed market prices. Next, as in the previous examples, we obtain a Monte Carlo sample of (S^*, V) , whose results are displayed in Figure 5. The mean and standard deviation of these simulated stock prices are $\bar{S}^* = 1.001237$ and $sd(S^*) = 0.2079399$, respectively. As can be seen, the implied Heston's RND is **positively skewed** ($sk = 1.027201$), as expected. Accordingly, we consider the distributions in Table 1 as possible RND candidates in this situation.

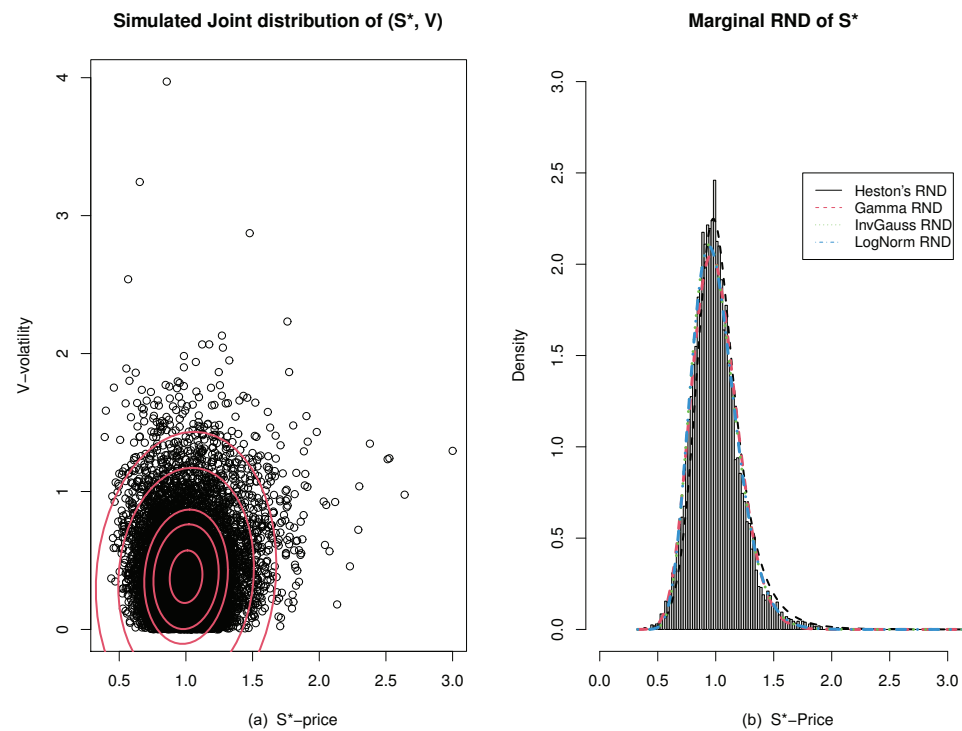


Figure 5. The simulated joint (S^* , V) distribution and Heston's, gamma, log-normal, and inverse Gaussian RNDs, calculated based on the AMD option data in Table 2.

Table 2. A comparison of the option prices obtained using the Heston's, gamma, inverse Gaussian, and Black–Scholes models for the 19 February 2021 option series of AMD (with 47 DTE), as of the closing of trading on 31 December 2020. The start to end time for calculating the option prices in this table was 0.1007299 s.

	MSE	0.004410	0.032725	0.018126	0.016748
Strike	Market Price	Heston's	Gamma	Inverse Gaussian	Black–Scholes
40.0	51.775	51.720	51.738	51.737	51.737
42.5	49.275	49.222	49.239	49.238	49.238
45.0	46.775	46.726	46.741	46.739	46.739
47.5	44.200	44.231	44.244	44.240	44.240
50.0	41.825	41.741	41.751	41.742	41.743
55.0	36.875	36.779	36.783	36.758	36.762
60.0	31.950	31.869	31.871	31.816	31.824
65.0	27.150	27.058	27.073	26.977	26.993
70.0	22.450	22.423	22.476	22.339	22.365
72.5	20.200	20.203	20.285	20.132	20.164
75.0	17.975	18.070	18.184	18.022	18.059
77.5	16.025	16.039	16.186	16.022	16.064
80.0	14.050	14.127	14.302	14.145	14.190
82.5	12.250	12.349	12.544	12.399	12.449
85.0	10.800	10.719	10.917	10.793	10.846
87.5	9.275	9.242	9.428	9.330	9.385
90.0	7.925	7.923	8.077	8.010	8.066

Table 2. Cont.

	MSE	0.004410	0.032725	0.018126	0.016748
Strike	Market Price	Heston's	Gamma	Inverse Gaussian	Black–Scholes
92.5	6.850	6.760	6.866	6.830	6.888
95.0	5.800	5.746	5.790	5.786	5.843
97.5	4.925	4.871	4.844	4.870	4.927
100.0	4.100	4.120	4.021	4.073	4.129
105.0	2.835	2.939	2.706	2.799	2.852
110.0	2.065	2.096	1.766	1.880	1.928
115.0	1.410	1.498	1.119	1.237	1.278
120.0	1.025	1.075	0.688	0.798	0.833
125.0	0.765	0.776	0.412	0.506	0.535
130.0	0.605	0.563	0.240	0.315	0.338
135.0	0.550	0.411	0.136	0.194	0.211
140.0	0.205	0.302	0.076	0.117	0.131
145.0	0.265	0.224	0.041	0.070	0.080
150.0	0.215	0.167	0.022	0.042	0.049
155.0	0.185	0.125	0.011	0.024	0.029
160.0	0.110	0.094	0.006	0.014	0.017
165.0	0.135	0.072	0.003	0.008	0.010
170.0	0.120	0.055	0.001	0.005	0.006
175.0	0.135	0.042	0.001	0.003	0.004
180.0	0.095	0.032	0.000	0.001	0.002
185.0	0.070	0.025	0.000	0.001	0.001
190.0	0.040	0.020	0.000	0.000	0.001

The source for the option market prices was Yahoo Financial: [www.https://finance.yahoo.com/](https://finance.yahoo.com/) (accessed on 1 January 2021).

Since in this case, we have the actual market option prices available, we can estimate the parameter $\nu = \sigma\sqrt{t}$, thereby defining these distributions directly using the ‘standard’ Black–Scholes implied volatility, namely $\hat{\nu} = IV^{BS}\sqrt{t}$. This entails using the `optim(.)` function again to minimize the $MSE(BS)$ with respect to the single parameter σ . This standard estimation procedure yields $IV^{BS} = 0.550085$, so that $\hat{\nu} = 0.1978301$. For this value, we present the graphs of the *gamma*, *inverse Gaussian*, and *log-normal* RNDs in Figure 5b, as in Table 1. The extent of their agreement with the Heston’s implied RND is evident. To further demonstrate this point, we calculate the option prices under each of the modeled RNDs and calculate the corresponding $MSE(Model)$. The results of this comparison are presented in Table 2. Finally, Figure 6 shows the option price curves for each of these pricing models, which are virtually almost identical in this example.

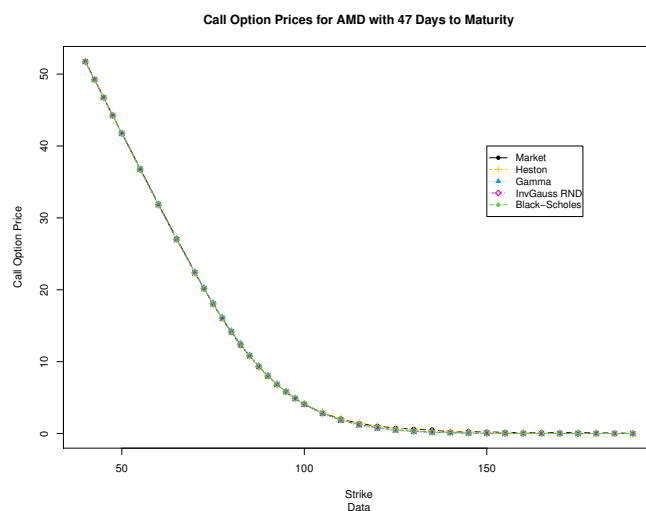


Figure 6. A comparison of the option prices obtained using the Heston's, gamma, inverse Gaussian, and Black–Scholes models for the 19 February 2021 option series of AMD (with 47 DTE), as of the closing of trading on 31 December 2020.

6. Concluding Remarks

It should be clear that the main results of this paper are not concerned with the estimation of the Heston's SV model parameters. Rather, this paper provides a theoretical solution for a long-standing 'open problem', namely, which class of risk neutral distributions, if any, satisfies Heston's stochastic volatility model for option valuation. In Section 3, we provided an affirmative theoretical answer to this question by characterizing the *unique* class of RNDs that corresponds to the SV model (1). In Theorem 2, we established that any RND for Heston's SV model *must* be a member of the class of scale parameter distributions that satisfies Assumption A, with the mean being the forward spot price. Indeed, any member of this class of scale parameter distributions could serve as a possible RND for the **direct** risk neutral calculation of an option price, as in (7) or Corollary 1. Each such RND allows for direct option valuation under Heston's SV model without the need to actually 'calibrate' or 'estimate' its five parameters. We pointed out that the particular choice of RND from within this class could be guided by the perceived skewness (i.e., an RND with an inherently negative skew could be used in the case of pricing options on an index, such as S&P500).

In Section 4, we identified and listed several well-known distributions that could serve as RNDs under Heston's SV model. These possible Heston's RNDs included one-parameter versions of the gamma, inverse gamma, Weibull, inverse Weibull, log-normal, and inverse Gaussian distributions. In a separate paper [17], we highlighted and detailed the usefulness of a two-parameter generalization of these distributions as an RND for Heston's SV model, namely, the generalized gamma (and its inverse) distribution. In Section 5, we illustrated our theoretical results in cases involving negatively skewed RNDs (i.e., the S&P500 and ODAX examples) and positively skewed RNDs (i.e., the AMD example). In the latter example, we used the **standard** MSE-based approach (see, for example, [6]) to illustrate the 'calibration' (estimation) of the five parameters of Heston's model and the single parameter of the comparable RNDs (i.e., the gamma, inverse Gaussian, and log-normal distributions). We then proceeded to calculate the price option chains accordingly. As can be seen from Table 2, the option prices calculated under the four 'competing' models were virtually identical, which well illustrates the main theoretical results of our paper (i.e., Theorem 2).

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Appendix A. Heston's (1993) Solution

In [1] Heston provided (semi)-closed-form expressions for the probabilities P_1 and P_2 , which comprise the solution $C_S(K)$ in (5) for option valuation under the stochastic volatility model in (1). Starting from a 'guess' of the Black–Scholes style solution:

$$C = SP_1 - Ke^{-rt}P_2, \quad (\text{A1})$$

he showed that with $x := \log(S)$, this solution must satisfy the SDE resulting from the SV model in (1)

$$\frac{\partial P_j}{\partial t} = \frac{1}{2}v \frac{\partial^2 P_j}{\partial x^2} + \rho\eta v \frac{\partial^2 P_j}{\partial x \partial v} + \frac{1}{2}\eta^2 v \frac{\partial^2 P_j}{\partial v^2} + (r + u_j v) \frac{\partial P_j}{\partial x} + (a - b_j v) \frac{\partial P_j}{\partial v},$$

for $j = 1, 2$, where $u_1 = 1/2$, $u_2 = -1/2$, $b_1 = \kappa - \rho\eta$, $b_2 = \kappa$, and $a = \kappa\theta$. These closed-form expressions are given by

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \mathcal{R}e \left[\frac{e^{-i\omega k} \psi_j(\omega, t, v, x)}{i\omega} \right] d\omega, \quad (\text{A2})$$

where $k := \log(K)$ and $\psi_j(\cdot)$ are the characteristic function

$$\psi_j(\omega, t, v, x) := \int_{-\infty}^\infty e^{i\omega s} p_j(s) ds \equiv e^{B_j(\omega, t) + D_j(\omega, t)v + i\omega x + i\omega r t}, \quad (\text{A3})$$

where $p_j(\cdot)$ is the *pdf* of $s_T = \log(S_T)$ corresponding to the probability P_j , $j = 1, 2$ and

$$B_j(\omega, t) = \frac{\kappa\theta}{\eta^2} \left\{ (b_j + d_j - i\omega\rho\eta)t - 2 \log\left(\frac{1 - g_j e^{d_j t}}{1 - g_j}\right) \right\}$$

$$D_j(\omega, t) = \frac{b_j + d_j - i\omega\rho\eta}{\eta^2} \left(\frac{1 - e^{d_j t}}{1 - g_j e^{d_j t}} \right)$$

$$g_j = \frac{b_j - i\omega\rho\eta + d_j}{b_j - i\omega\rho\eta - d_j}$$

$$d_j = \sqrt{(i\omega\rho\eta - b_j)^2 - \eta^2(2i\omega u_j - \omega^2)}.$$

We point out that d_j is taken to be the positive root of the Riccati equation involving D_j . However, using the negative root instead, namely $d'_j = -d_j$, has been shown to provide an equivalent but more stable solution for ψ_j (see [29] for more details about this so-called 'Heston Trap'). In either case, efficient numerical routines, such as the `cfHeston` and `callHestoncf` functions of the NMOF package of R, [30], are readily available to accurately compute the values of ψ_j and hence, the values of P_j and call option values for given t, s , and v values, as well as any choice of $\theta = (\kappa, \theta, \eta, \rho)$.

Now, having established (A3), the standard application of the Fourier transform shows (see, for example, [31]) that the *pdf* $p_j(\cdot)$ of $s_T = \log(S_T)$ can be obtained for $s \in \mathbb{R}$:

$$p_j(s) = \frac{1}{\pi} \int_0^\infty \mathcal{R}e \left[e^{-i\omega s} \psi_j(\omega, t, v, x) \right] d\omega. \quad (\text{A4})$$

Hence, it follows immediately that the *pdf* $\tilde{p}_j(\cdot)$ of S_T is given by

$$\tilde{p}_j(u) = \frac{1}{u} \times p_j(\log(u)) \equiv \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-i\omega \log(u)} \psi_j(\omega, t, v, x)}{u} \right] d\omega, \quad u > 0.$$

Further, since the characteristic functions ψ_j in (A3) are affine in $x + rt = \log(S) + rt \equiv \log(\mu)$, where $\mu = Se^{rt}$, as in Corollary 1, we can rewrite $\tilde{p}_j(u)$ as

$$\tilde{p}_j(u) = \frac{1}{\mu\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-i\omega \log(u/\mu)} \tilde{\psi}_j(\omega, t, v)}{u/\mu} \right] d\omega, \quad (\text{A5})$$

where

$$\log(\tilde{\psi}_j(\omega, t, v)) := \log(\psi_j(\omega, t, v, x) - i\omega x - i\omega rt).$$

We point out that in light of (8), the $\tilde{p}_2(\cdot)$ in (A5) is the RND (under $\mathbb{Q}$) for Heston's model and can similarly be easily evaluated numerically alongside the evaluation of P_2 . Indeed, we have

$$P_2 \equiv \int_K^\infty \tilde{p}_2(u) du = \mathbb{Q}(S_T > K).$$

It should be noticed from expression (A5) that any $\tilde{p}_2(\cdot)$ of a Heston's model RND and the corresponding risk neutral distribution $Q_\mu(\cdot)$ of S_T constitute a scale family of distributions in $\mu = Se^{rt}$, so that it satisfies the terms of Assumption A. This assertion is summarized in Proposition A1.

Proposition A1. Let $q_\mu(\cdot)$ be any RND with a corresponding risk neutral distribution $Q_\mu(\cdot)$ that satisfies Heston's solution in (5) with $\mu = Se^{rt}$. Then, $q_\mu(\cdot)$ is in the form given in (A5) and, therefore, $Q_\mu(\cdot)$ must be a member of a scale family of distributions in μ .

The result stated in the next claim is known but the details are instructive to proving (11).

Claim 1. Let $\Delta(K) = \partial C / \partial S$, as in (10). Then, for the Heston's solution in (5) (or (A1)) with P_j , $j = 1, 2$, as given in (A2), we have $\Delta(K) = P_1$.

Proof. By (A1), we have

$$\frac{\partial C}{\partial S} = P_1 + S \frac{\partial P_1}{\partial S} - Ke^{-rt} \frac{\partial P_2}{\partial S}. \quad (\text{A6})$$

Since we have $x = \log(S)$, it follows from (A2) that

$$\frac{\partial P_j}{\partial S} = \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-i\omega k} \psi_j(\omega, t, v, x)}{S} \right] d\omega.$$

Hence, by (A4) and with $\mu = Se^{rt}$, we have

$$S \frac{\partial P_j}{\partial S} \equiv p_1(k), \quad \text{and} \quad Ke^{-rt} \frac{\partial P_2}{\partial S} \equiv \frac{K}{\mu} p_2(k),$$

which, by (9), implies that $S \partial P_1 / \partial S - Ke^{-rt} \partial P_2 / \partial S = 0$ in (A6). $\square$

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